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WO-400-02 — Overhaul pump P-103B (mechanical seal and bearings)

Area	A400 Pumps	**Task**	PO Pump overhaul
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	P-103B — P-103B centrifugal pump — E 179'-0" N -18'-6" EL 100'-9"		
Equipment	P-103B	**Lines**	4"-P300-P-1014, 3"-P300-P-1016, 3/4"-P150-W-3022
Isometrics	ISO-P-1014, ISO-P-1016, ISO-W-3022 (owner's drawings, attached)		
Permits	General work permit, Lifting plan		

Reason

High bearing vibration; overhaul seal and bearings.

Scope of work

1. Isolate P-103B electrically and per the isolation list; drain the casing.
2. Record as-found shaft alignment; remove the coupling guard and spacer.
3. Break the suction and discharge nozzle joints; remove the pump to the workshop.
4. Replace the mechanical seal and the bearings; inspect the impeller and wear rings.
5. Reinstall; new nozzle gaskets and studs tightened per SW-WO-5; align to the owner's alignment criteria.
6. Remove blinds and isolations; run test.

Affected items: P-103B

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1014	4"-P300-P-1014	4"-P300-P-1014	1	0 error(s), 1 warning(s) — see below
ISO-P-1016	3"-P300-P-1016	3"-P300-P-1016	1	1 error(s), 0 warning(s) — see below
ISO-W-3022	3/4"-P150-W-3022	3/4"-P150-W-3022	1	0 error(s), 1 warning(s) — see below

ISO-P-1014 (PDF): ../isometrics-imported/ISO-P-1014.pdf · ISO-P-1016 (PDF): ../isometrics-imported/ISO-P-1016.pdf · ISO-W-3022 (PDF): ../isometrics-imported/ISO-W-3022.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1014	1	warning	branch weld 3/4"-P300-I-7035/W1 (E 176'-6" N -11'-0" EL 102'-3 5/16", sheet 1) has no label
ISO-P-1016	1	error	sheet 1: CONT'D ON ISO-P-1015 SH 1 prints E 174'-2 1/2" N -17'-6" EL 107'-6" for E 174'-2 1/2" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 3"-P300-P-1016 meets 3"-P300-P-1015 (E 174'-2 1/2" N -16'-6" EL 107'-6")
ISO-W-3022	1	warning	weld 3/4"-P150-W-3022/W8 (E 181'-0" N -16'-6" EL 124'-10 3/4", sheet 1) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-103B	equipment	P-103B centrifugal pump		E 179'-0" N -18'-6" EL 100'-9"
P-103B-M	motor	Motor of P-103B		E 179'-0" N -18'-6" EL 100'-9"
4"-P300-P-1014/B7	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1014	E 179'-0" N -13'-3-11/16" EL 102'-0"
4"-P300-P-1014/B5	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1014	E 179'-0" N -14'-2-1/8" EL 102'-0"
3"-P300-P-1016/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1016	E 179'-0" N -16'-6" EL 105'-2-3/16"
3"-P300-P-1016/A2	item	VALVE, SWING CHECK 3" P300	3"-P300-P-1016	E 179'-0" N -16'-6" EL 104'-1/8"
3/4"-P150-W-3022/A3	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-W-3022	E 178'-0" N -4'-0" EL 123'-7-11/16"
3/4"-P150-W-3022/A6	item	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3/4"-P150-W-3022	E 178'-0" N -4'-0" EL 124'-4-5/8"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	electrical	P-103B-M		Motor of P-103B: isolated, locked and tried at the starter			
2	valve	4"-P300-P-1014/B7	4"-P300-P-1014	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1014/B7) on P-103B-S	E 179'-0" N -13'-3-11/16" EL 102'-0"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
3	blind	4"-P300-P-1014/B5	4"-P300-P-1014	Fit a blind between 4"-P300-P-1014/B7 and 4"-P300-P-1014/B5, P-103B side	E 179'-0" N -14'-4-7/8" EL 102'-0"		
4	valve	3"-P300-P-1016/A5	3"-P300-P-1016	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1016/A5) on P-103B-D	E 179'-0" N -16'-6" EL 105'-2-3/16"		
5	blind	3"-P300-P-1016/A2	3"-P300-P-1016	Fit a blind between 3"-P300-P-1016/A5 and 3"-P300-P-1016/A2, P-103B side	E 179'-0" N -16'-6" EL 103'-5"		
6	valve	3/4"-P150-W-3022/A3	3/4"-P150-W-3022	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-W-3022/A3) on P-103B-F	E 178'-0" N -4'-0" EL 123'-7-11/16"		
7	blind	3/4"-P150-W-3022/A6	3/4"-P150-W-3022	Fit a blind between 3/4"-P150-W-3022/A3 and 3/4"-P150-W-3022/A6, P-103B side	E 178'-0" N -4'-0" EL 124'-2-3/8"		

Parts

Description	Qty	Replaces
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1014/B1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1014/B3
GASKET 3" P300 COMPRESSED SHEET	1	3"-P300-P-1016/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	3"-P300-P-1016/A3
Mechanical seal kit for P-103B	1	
Bearing set for P-103B	1	

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1014/B1	4 × 1" studs (4"-P300-P-1014/B3)	35,343	3	
3"-P300-P-1016/A1	4 × 0.875" studs (3"-P300-P-1016/A3)	27,059	3	

Test — Run test

Description	Running at duty: no seal leak; bearing temperature and vibration within the owner's limits
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

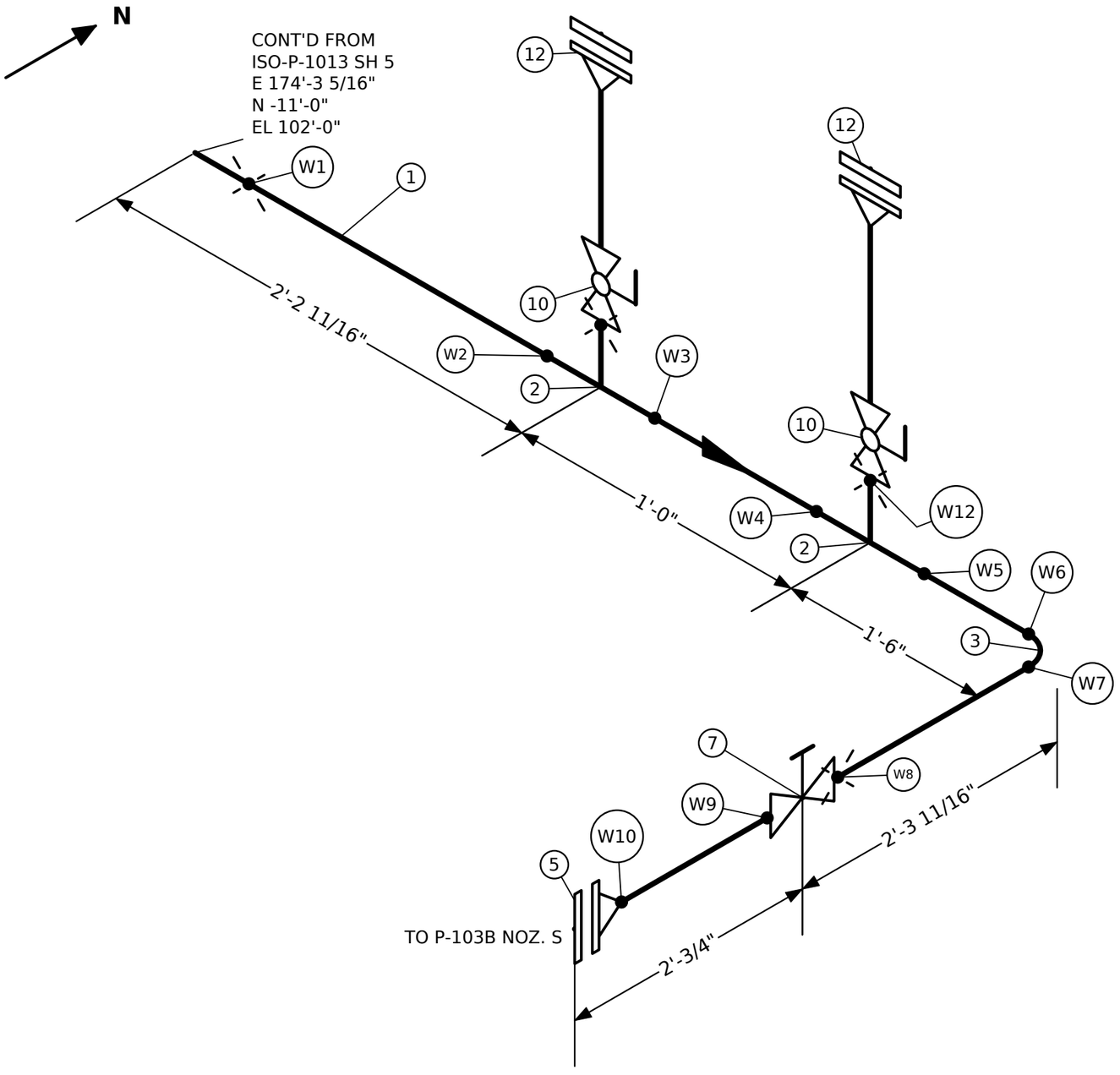
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	As-found alignment recorded	Owner alignment procedure	Recorded	Alignment record	W	R	—				
3	Seal and bearings replaced; impeller and wear rings inspected	Owner pump procedure	Clearances within the owner's criteria	Pump overhaul record	S	R	—				
4	Nozzle joints remade with new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
5	Final alignment	Owner alignment procedure	Within the owner's alignment criteria	Alignment record	W	W	—				
6	Run test at duty	SW-TEST-2	No seal leak; temperatures and vibration within owner limits	Run test record	W	H	—				
7	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
8	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1014/B7	isolation: valve	ISO-P-1014	1	
4"-P300-P-1014/B5	isolation: blind	ISO-P-1014	1	
3"-P300-P-1016/A5	isolation: valve	ISO-P-1016	1	
3"-P300-P-1016/A2	isolation: blind	ISO-P-1016	1	
3/4"-P150-W-3022/A3	isolation: valve	ISO-W-3022	1	
3/4"-P150-W-3022/A6	isolation: blind	ISO-W-3022	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	4'-2 1/2"	4"	PIPE 4" P300 BEVELLED ENDS
2	2	4"x3/4"	TEE REDUCING 4" × 3/4" P300
3	1	4"	ELBOW 90° R = 5/3 OD 4" P300
4	3	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
5	3	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
6	3	4"	GASKET 4" P300 COMPRESSED SHEET
7	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
8	2	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
9	4	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
10	2	3/4"	VALVE, BALL, FLOATING 3/4" P300
11	4	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
12	2	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

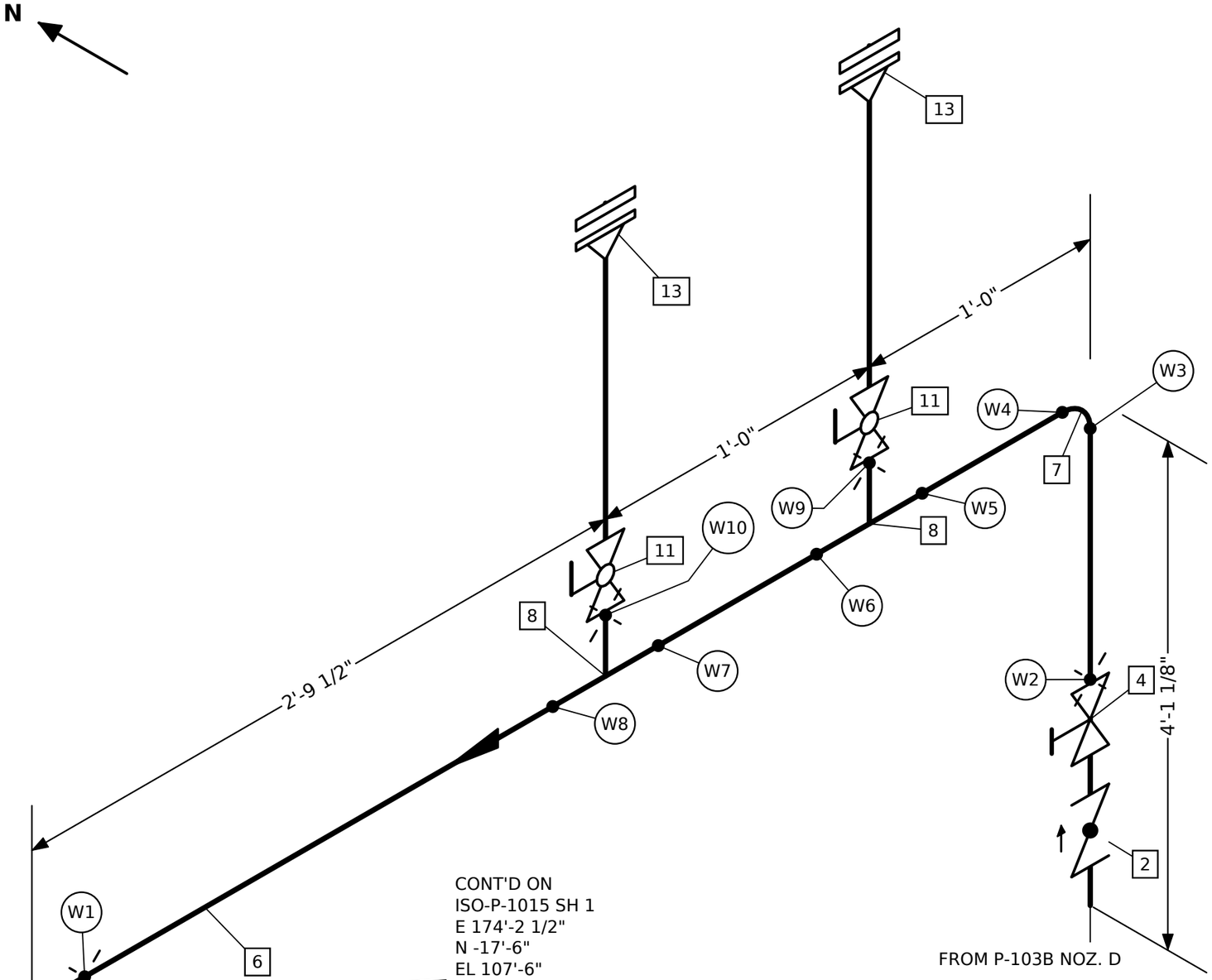
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1014	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1014	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	3"	GASKET 3" P300 COMPRESSED SHEET
2	1	3"	VALVE, SWING CHECK 3" P300
3	3	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300
4	1	3"	VALVE, GATE, WEDGE, OS&Y 3" P300
5	1	3"	FLANGE WELD-NECK 3" P300 RAISED FACE
6	4'-5 5/16"	3"	PIPE 3" P300 BEVELLED ENDS
7	1	3"	ELBOW 90° R = 5/3 OD 3" P300
8	2	3"x3/4"	TEE REDUCING 3" × 3/4" P300
9	2	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
10	4	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
11	2	3/4"	VALVE, BALL, FLOATING 3/4" P300
12	4	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
13	2	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

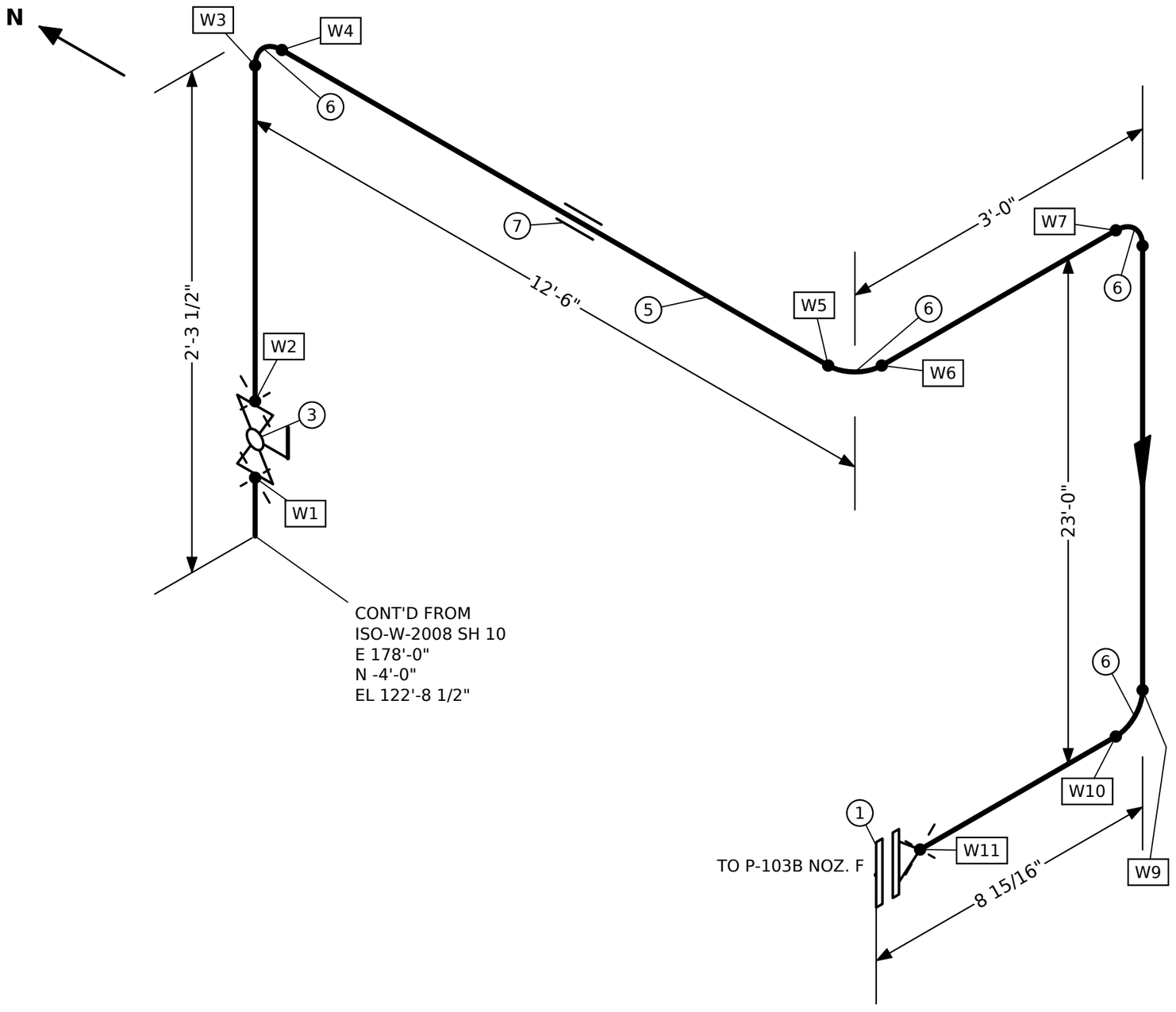
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3"-P300-P-1016	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1016	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	3/4"	FLANGE WELD-NECK 3/4" P150 RAISED FACE
2	3	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
3	1	3/4"	VALVE, BALL, FLOATING 3/4" P150
4	3	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
5	38'-5 7/16"	3/4"	PIPE 3/4" P150 BEVELLED ENDS
6	4	3/4"	ELBOW 90° R = 5/3 OD 3/4" P150
7	1	3/4"	HANGER FOR 3/4" LINE (HANGER-0.75x88.0)

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

WATER (COOLING AND PROCESS)

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	W

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P150-W-3022	W150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-W-3022	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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